Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

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```c
#include <stdio.h>

int countZeroes(int a[], int left, int right) {
    if (left > right)
        return 0;

    int mid = (left + right) / 2;

    if (a[mid] == 0)
        return (right - mid + 1) + countZeroes(a, left, mid - 1);

    return countZeroes(a, mid + 1, right);
}

int main() {
    int m;
    scanf("%d", &m);

    int a[m];
    int i = 0;

    while (i < m) {
        scanf("%d", &a[i]);
        i = i + 1;
    }

    printf("%d", countZeroes(a, 0, m - 1));

    return 0;
}
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔

Open block draw

✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.